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Research article

Information systems auditing of psychiatric diagnoses using context-enriched QA-based language models



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ABSTRACT

The growing use of artificial intelligence (AI)-driven decision support systems (DSS) in healthcare requires effective auditing frameworks to ensure diagnostic accuracy and transparency. This study introduces a novel information system (IS) approach for psychiatric diagnosis auditing, transforming traditional multi-label classification into a question-answering (QA)-based framework enriched with diagnostic descriptions and explanations. Evaluating traditional classifiers, transformer models, large language model (LLM), and the proposed psychiatric QA model, results show that task transformation and context enrichment improve accuracy, with the QA model achieving the highest sum-F1-score and lowest precision fail rate. This study contributes to AI-driven IS, algorithmic auditing, and healthcare decision support, with managerial implications for electronic health record integration, insurance claim validation, and clinician workflow optimization.

1. Introduction

As demonstrated by Aminabee (2024) and Stein et al. (2022), the accuracy of psychiatric diagnoses is critical for effective treatment planning, patient outcomes, and healthcare cost management. However, psychiatric disorders often present with overlapping symptoms and subjective diagnostic criteria, making them prone to misdiagnosis, inconsistencies across clinicians, and insurance disputes (Allsopp et al., 2019; Berner and Graber, 2008; Chaturvedi, 2025; Clark et al., 2017). Unlike other medical fields where diagnostic imaging or laboratory tests provide objective assessments, psychiatric evaluations rely heavily on clinical judgment and textual documentation, leading to significant challenges in artificial intelligence (AI)-driven auditing frameworks for decision support systems (DSS) in information system (IS) (Abdel-Karim et al., 2023; Berner and Graber, 2008; Sharma and Kaushik, 2025). The absence of standardized IS-driven auditing mechanisms for psychiatric diagnoses can result in treatment errors, improper insurance claims, and increased healthcare costs (Sbodio et al., 2024; Xing, 2025).

Because psychiatric documentation is narrative and clinician judgments are subjective, the diagnostic labels written into electronic health records (EHRs) are often noisy; once stored, they quickly become de facto "ground truth" that feeds decision support, quality reporting, and reimbursement, allowing errors to propagate silently across clinical and administrative workflows. Organizations therefore require a scalable, context-aware auditing mechanism that links each recorded label to the evidence in the note and that explicitly operates under label noise and overlapping symptom profiles. We frame this as an IS auditing problem: making diagnostic consistency, subgroup fairness, and safety-critical performance observable and actionable, thereby strengthening transparency and trust in AI-enabled DSS and laying the groundwork for the responsible use of AI in psychiatric care.

The emergence of AI-driven DSS has transformed clinical auditing and risk assessment, offering the potential to enhance consistency and transparency in psychiatric evaluations (Burton-Jones and Volkoff, 2017; Bzdok and Meyer-Lindenberg, 2018; Chekroud et al., 2021; Dwyer et al., 2018). Although machine learning (ML) models have demonstrated success in medical diagnostics, existing approaches primarily focus on predicting psychiatric diagnoses rather than auditing clinician-assigned labels for accuracy and consistency (Lee et al., 2023; Sun et al., 2023). Furthermore, traditional multi-label classification models struggle with psychiatric diagnosis auditing due to label ambiguity, symptom overlap, and lack of interpretability (Abd-Alrazaq et al.,

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2023; Gibson et al., 2022; Omar et al., 2024). Despite advancements in AI-based medical diagnostics, there is no established auditing framework for psychiatric assessments driven by AI. This study fills this gap by introducing a structured IS-driven question-answering (QA)-based auditing system for psychiatric diagnoses.

To address these challenges, we proposed a novel AI-based psychiatric auditing framework that reconfigures psychiatric diagnosis validation as a QA task. Unlike traditional multi-label classification models that independently assign diagnostic labels, our approach leverages context-enriched QA-based auditing, allowing AI models to interpret clinical notes within the context of structured diagnostic knowledge. Specifically, this study introduced the following hypotheses:

(1) Task transformation: Reformulating psychiatric diagnosis auditing as a QA-based problem, allowing for context-aware, flexible decision-making rather than rigid classification.
(2) Context enrichment: Enhancing AI auditing with diagnostic descriptions and explanations, improving model interpretability and reducing errors.
(3) Risk categorization framework: Introducing a structured risk assessment model to systematically classify diagnoses into low, medium, or high-risk categories, ensuring greater transparency in AI-assisted psychiatric auditing.

This study makes several contributions to IS research and AI-driven auditing. First, it extends the role of AI in DSS by demonstrating how task transformation and context enrichment can enhance algorithmic auditing frameworks for psychiatric evaluations. Second, it provides a novel AI-driven risk assessment approach that offers a structured mechanism for identifying and evaluating diagnostic inconsistencies. Third, this study contributes to healthcare IS research by proposing a scalable AI-assisted auditing framework that can be integrated into EHRs and insurance claim validation systems.

Beyond theoretical advancements, this study has practical implications for hospitals, insurance firms, and regulatory bodies seeking to reduce diagnostic errors, improve insurance claim accuracy, and optimize AI-assisted medical auditing. Future research should explore multi-institutional validation, human-in-the-loop auditing, and scalable IS deployment strategies to ensure robust AI adoption for psychiatric quality assurance.

2. Theoretical background and literature review

As emphasized by Bagheri et al. (2025), ensuring psychiatric diagnostic accuracy is not only vital for patient care but is also critical in healthcare DSS, insurance claim validation, and AI-driven auditing frameworks in IS. However, psychiatric assessments rely on qualitative evaluations rather than objective biomarkers, making them prone to misdiagnosis, inconsistencies, and subjectivity. Traditional medical diagnostics benefit from structured risk assessment models, as discussed by Wang et al. (2023) and Evans et al. (2016), but psychiatric diagnosis auditing lacks standardized AI-driven frameworks to ensure accuracy, consistency, and fairness. This section reviews the literature on risk assessment in medical diagnostics, the role of ML in healthcare, challenges in psychiatric diagnostic accuracy, and the existing gaps in AI-based psychiatric auditing.

Ensuring psychiatric diagnostic accuracy is vital for patient care and, from an IS perspective, for dependable decision support, reimbursement, and governance workflows (Bagheri et al., 2025; Wu et al., 2022). In contrast to objective, test-based specialties, psychiatric assessments are primarily text- and judgment-driven, which amplifies label noise and downstream propagation of errors across DSS, billing, and quality reporting. In this section, we focus on AI-driven auditing, post-hoc verification, and triage of clinician-assigned labels using computational methods rather than diagnostic prediction.

2.1. Risk assessment in medical diagnostics

Risk assessment is a cornerstone of medical diagnostics, providing clinicians with tools to evaluate patient conditions and guide treatment decisions (Jayamani et al., 2022; McGowan et al., 2023; Njoroge and Nichols, 2014). Standardized tools such as the Framingham risk score for cardiovascular disease (Bosomworth, 2011) and the tumor, node, metastasis (TNM) staging system for cancer discussed by Rosen and Sapra (2022) and Saoud et al. (2020) illustrate the value of structured approaches for estimating health risks. These methods integrate demographic, clinical, and diagnostic data to categorize patients into at-risk groups, enabling proactive interventions.

However, psychiatric risk assessment presents unique challenges owing to its reliance on qualitative evaluations rather than objective biomarkers (García-Gutiérrez et al., 2020; McGorry et al., 2014). Reliance on patient narratives, behavioral observations, and structured interviews introduces variability, according to Marey et al. (2024) and Rutter (2011). This subjectivity can lead to diagnostic inconsistencies, potential biases, and missed risk factors, particularly when symptoms overlap or are ambiguous (Bradford et al., 2024; Meyer et al., 2021).

While traditional risk assessment models effectively structure decision-making for physical health conditions, they exhibit limitations when applied to psychiatric evaluation:

- Subjectivity and bias: Psychiatric evaluations can differ significantly across clinicians, as noted by Ten Cate and Regehr (2019).
- Oversimplification: Aftab et al., (2024) argued that existing diagnostic frameworks may fail to fully capture the complexity of mental health symptoms, leading to overly rigid diagnostic categories.
- Static nature: According to Wang et al. (2021), many assessments lack adaptability to dynamically evolving patient conditions.

These limitations underscore the need for an AI-driven DSS capable of auditing psychiatric diagnoses by analyzing unstructured clinical notes, detecting inconsistencies, and improving diagnostic transparency.

2.2. Machine learning (ML) in medical diagnostics

While risk assessment frameworks have improved decision-making in structured medical contexts, psychiatric diagnosis remains uniquely challenging due to its inherent subjectivity, necessitating AI-driven innovations in IS and healthcare (Dakanalis et al., 2024; Guo et al., 2024; Thakkar et al., 2024). ML has emerged as a transformative force in medical diagnostics, enabling the automated interpretation of large datasets, identifying latent patterns, and supporting clinician decision-making (Javaid et al., 2022; Jussupow et al., 2021; Zhang et al., 2022).

Early ML applications in healthcare primarily focused on structured data tasks such as image recognition in radiology and pathology (Li et al., 2022). Convolutional neural networks (CNNs) have demonstrated expert-level performance in identifying anomalies in medical imaging (Kourounis et al., 2023; Malik et al., 2020), whereas other ML techniques have been widely used for predictive analytics—forecasting patient readmissions, disease progression, and treatment outcomes (Badawy et al., 2023; Park et al., 2021).

In psychiatric diagnostics, ML models have introduced several promising capabilities:

- Predictive modeling: As shown by Chekroud et al. (2021), algorithms forecast psychiatric conditions based on patient history, enabling early intervention strategies.
- Natural language processing: Jackson et al. (2017) demonstrated that AI extracts insights from unstructured psychiatric notes and identifies subtle symptom patterns.

- Automated screening: According to Aggarwal et al. (2023), AI-powered tools conduct preliminary mental health assessments and identify at-risk patients.
- Personalized treatment: ML tailors treatment plans based on patient-specific profiles, improving therapeutic outcomes, as highlighted by Adam et al. (2020).

While these advancements demonstrate ML's success in predictive analytics, its potential for auditing psychiatric diagnoses remains largely unexplored. The absence of AI-driven auditing frameworks means that misdiagnoses persist, leading to inappropriate treatments and inflated healthcare costs (Berner and Graber, 2008; Chaturvedi, 2025).

This study builds upon existing ML research but reframes the problem from diagnosis prediction to AI-driven auditing—an approach that has received little attention in IS literature.

2.3. Challenges in psychiatric diagnostic accuracy

Psychiatric diagnoses are particularly vulnerable to inaccuracies due to several unique factors:

(1) Subjective symptom evaluation: Unlike most physical conditions, mental disorders lack definitive biomarkers and rely on self-reported symptoms, making diagnoses open to interpretation biases (Konjevod et al., 2025).
(2) Symptom overlap: Many psychiatric conditions share overlapping features (e.g., depression and anxiety), complicating differential diagnosis (Vasconcelos-Moreno et al., 2025).
(3) Comorbid conditions: Patients frequently experience multiple co-occurring mental health disorders, making diagnostic precision difficult (Keramatian and Levit, 2023).
(4) Evolving diagnostic criteria: Updates to classification systems such as the Diagnostic and Statistical Manual of Mental Disorders and the International Classification of Diseases (ICD) create inconsistencies in diagnostic standards across clinicians (Rix, 2026).
(5) Cultural and social factors: Cultural norms influence both symptom expression and clinician interpretation, adding complexity to psychiatric assessment (Efremov, 2025).

These challenges highlight the urgent need for AI-assisted auditing to validate clinician diagnoses by cross-referencing psychiatric notes with diagnostic criteria and identifying inconsistencies.

2.4. Research gaps and contribution rationale

Despite the increasing interest in ML-driven psychiatric diagnostics, several critical research gaps remain.

- Prediction vs. auditing: As Chekroud et al. (2021) and Dwyer et al. (2018) point out, most studies focus on predicting psychiatric conditions rather than evaluating the validity of clinician-assigned labels.
- Handling label noise: Training ML models on misdiagnosed patient records reinforces errors (Bordini et al., 2017), yet few studies explicitly address label noise in psychiatric datasets.
- Lack of context-aware AI models: According to Marey et al. (2024) and Amirahmadi et al. (2023), existing ML models operate as black boxes and fail to incorporate contextual patient narratives.
- Task reformulation in psychiatric auditing: Although task reformulation techniques (e.g., multi-label classification as QA tasks) have been explored in other domains (Dai et al., 2021; Ren et al., 2020), their application in AI-driven psychiatric auditing is novel.
- Question-centric large language model (LLM) frameworks vs. auditing: Recent advances, such as the HealthQ framework (Wang et al., 2025), have begun to formalize question generation and evaluation in healthcare LLM chains. However, these approaches focus on assessing the quality of questions, such as their specificity, relevance, and usefulness, rather than auditing existing psychiatric diagnoses or verifying clinician-assigned labels, leaving a critical gap that this study addresses.

To address these gaps, this study introduces the following: (i) Task transformation: reframing multi-label diagnosis checking as QA, which operationalizes auditability and traceability in DSS; (ii) Context enrichment: injecting diagnostic descriptions to enhance transparency and reviewer sensemaking within IS workflows; and (iii) risk categorization for triage: structuring outputs into low/medium/high tiers that instantiate actionability for governance. Together, these contributions move beyond accuracy gains to provide an IS-grounded auditing mechanism that integrates EHR and claims processes while supporting human oversight.

3. Research methodology

To develop an AI-assisted auditing framework for psychiatric diagnoses, this study employed a computational experimental approach that leverages ML and natural language processing (NLP) techniques. The methodology followed a structured workflow comprising data collection and preparation, task transformation, model development and tuning, system integration, and evaluation. This approach ensures rigorous validation of the proposed context-enriched QA-based auditing system.

3.1. Data collection and preparation

Data for this study were collected from a psychiatric clinic, with a focus on extracting unstructured textual information from doctors' notes associated with each patient visit. The dataset comprised 48,492 patient records from 7,000 unique patients, including patient demographics, diagnosis codes, treatment details, medication information, and free-text clinical notes written by psychiatrists.

We converted the physical and digital patient records into machine-readable text using OCR for PDF sources, followed by post-OCR normalization (de-hyphenation, unicode normalization, and header/footer removal). Notes were then sectioned and processed using NLP routines to extract structured elements—ICD codes, medications, and treatment plans via pattern recognizers and curated lexicons. Regular expressions captured ICD patterns and medication dose templates, whereas rule-based negation handling suppressed spurious matches. Automated validation flagged ill-formed codes, duplicated sections, and temporal inconsistencies for targeted review, thereby improving the reliability of the resulting dataset.

Patient privacy and data security are prioritized. All personally identifiable information, such as names and contact details, were systematically removed from the clinical notes. Patient medical record numbers were encrypted and replaced with unique, anonymized identifiers. The anonymized dataset was stored on a secure, password-protected server, with access restricted to authorized research personnel, ensuring compliance with ethical and legal standards for data protection.

This cleaned and privacy-protected dataset served as the foundation for model training, evaluation, and comparison, forming the basis for our investigation of task transformation and context-enriched psychiatric diagnosis auditing.

3.2. Task transformation: multi-label classification to QA

Traditional ML models for psychiatric diagnosis typically frame the problem as a multi-label classification task, where each patient record is assigned multiple diagnostic codes based on clinical notes. However, this approach often struggles with overlapping symptoms,

ambiguous diagnostic boundaries, and label noise, particularly in psychiatry, where diagnoses frequently evolve based on clinician judgment. To address these limitations, this study reframes the diagnostic classification problem as a QA task by leveraging the LLM to generate diagnostic outputs in a flexible and context-aware manner.

Rather than treating diagnostic label auditing as a direct classification problem, the proposed QA-based model uses a doctor's clinical notes as input and answers a structured question about the patient's mental health conditions. The model was trained to generate diagnostic codes in response, mimicking a structured medical reasoning process. This task transformation aligns more closely with the way psychiatrists make diagnostic decisions, where reasoning is guided by patient narratives rather than fixed input-output mappings.

To implement this transformation, each patient record was reformulated into a structured QA format. The input consists of the doctor's notes combined with a query, such as: "what is the psychiatric diagnostic code?" The output is a sequence of ICD diagnostic codes with their descriptions, optionally augmented with brief explanations when context enrichment was applied. This approach allows the model to interpret unstructured clinical narratives dynamically, rather than relying solely on patterns learned from static label distributions. By generating diagnostic labels in response to an inquiry, the model can adapt to nuanced symptom presentations and infer missing or implicit diagnostic indicators.

This QA transformation also enables greater interpretability and mitigates errors stemming from label inconsistencies in training data. Since psychiatric diagnoses are often subject to variability in clinician judgment, a QA model can better accommodate diagnostic uncertainty by reasoning over the textual context instead of rigidly mapping notes to pre-defined labels. Moreover, this framing allows for the potential integration of explanatory reasoning steps, further enhancing the model's usability as an auditing and quality-assurance tool.

The transition from a multi-label classification paradigm to a QA-based model represents a fundamental shift in how psychiatric diagnostic auditing is approached. By leveraging this structured QA framework as shown in Fig. 1, the proposed methodology improves model flexibility, interpretability, and diagnostic accuracy, making it particularly well-suited for auditing existing psychiatric diagnoses in real-world clinical settings.

3.3. Context enrichment: description and explanation

Psychiatric diagnoses often involve complex, overlapping symptoms that require clinicians to differentiate between multiple, closely related conditions. For example, Mood [affective] disorders (F3) can exhibit similarities with anxiety, dissociative, stress-related, somatoform and other nonpsychotic mental disorders (F4), whereas schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders (F2) may share features with mood disorders. This diagnostic ambiguity poses challenges not only for human practitioners but also for ML models, particularly when the training data itself may contain inaccuracies or inconsistencies. To address this, we introduced context enrichment, which involves augmenting the input provided to the model with diagnostic descriptions and explanatory information tied to each code.

Context enrichment operates on the principle that supplementing clinical notes with external knowledge about diagnostic categories can improve the model's capacity to disambiguate overlapping conditions. In this study, two types of contextual information were integrated into the input:

- Diagnostic descriptions: Brief textual definitions of each psychiatric diagnosis, such as "F2: schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders."
- Explanations: exemplary links between clinical symptoms and diagnostic codes, such as "F2: schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders, schizophrenia, schizotypal disorder, delusional disorders, brief psychotic disorder, shared psychotic disorder, schizoaffective disorders, other psychotic disorder not due to a substance or known physiological condition, Unspecified psychosis not due to a substance or known physiological condition."

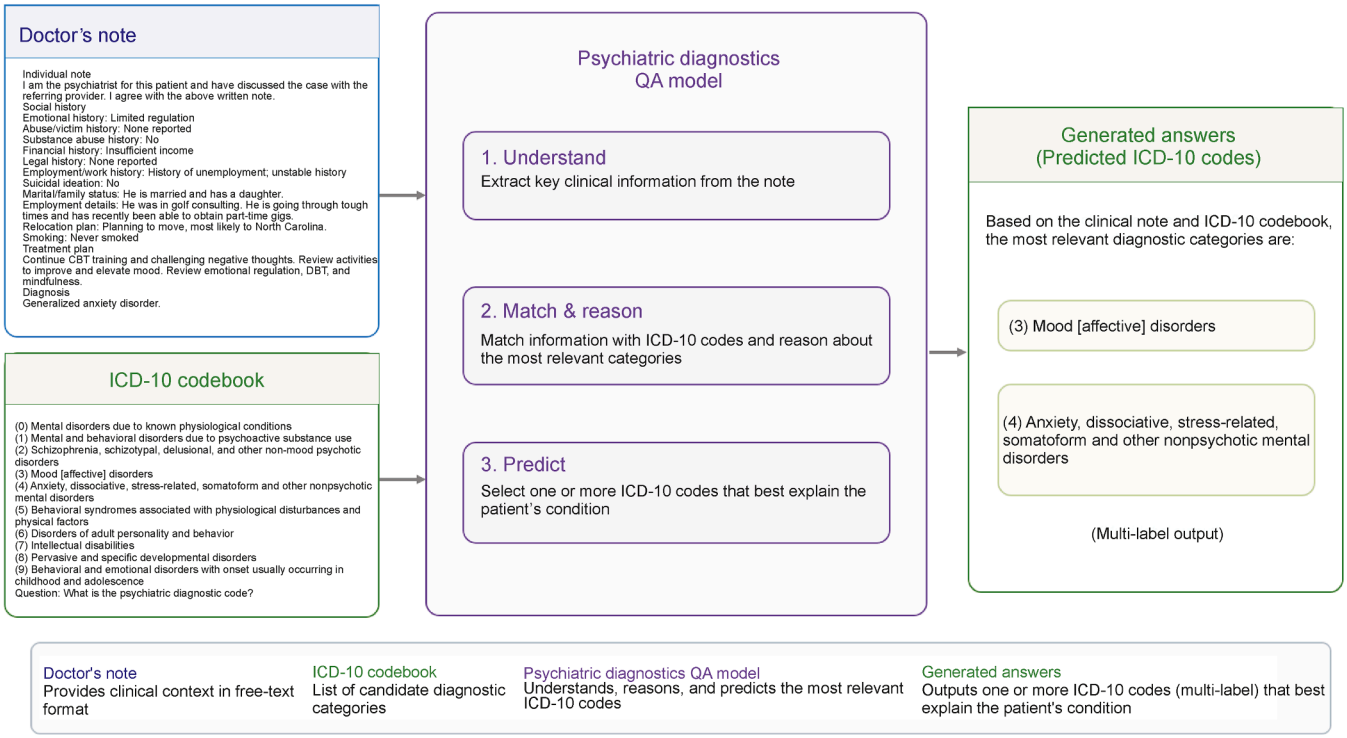


Fig. 1. Overview of the psychiatric diagnostics question-answering (QA) model.

Table 1
Comprehensive listing of diagnosis codes and corresponding descriptions.

Code	Description
F0	Mental disorders due to known physiological conditions
F1	Mental and behavioral disorders due to psychoactive substance use
F2	Schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders
F3	Mood [affective] disorders
F4	Anxiety, dissociative, stress-related, somatoform and other nonpsychotic mental disorders
F5	Behavioral syndromes associated with physiological disturbances and physical factors
F6	Disorders of adult personality and behavior
F7	Intellectual disabilities
F8	Pervasive and specific developmental disorders
F9	Behavioral and emotional disorders with onset usually occurring in childhood and adolescence

Table 1 presents a comprehensive list of diagnosis codes and their respective descriptions. This contextual information was appended to the clinical note before being processed by the model. The goal was to equip the model with diagnostic criteria during inference, mimicking the cognitive process by which clinicians cross-check patient symptoms against formal diagnostic guidelines. The enriched input helps the model to resolve ambiguous cases by relating patient symptoms to diagnostic descriptions, ultimately reducing the likelihood of misclassification.

We have attached two context types per code: diagnostic descriptions and explanations. Snippets are length-capped to maintain a fixed token budget across cases and are style-normalized for consistency.

At inference, each note is paired with a fixed-order inventory F0 → F9 and rendered with stable delimiters as follows:

```
[QUESTION]

Given the clinical note, what is the psychiatric diagnostic code?

Return any subset of codes from {F0,F1, …,F9} as a comma-separated list.

[NOTE]

{<sectioned clinical text;>}

[CONTEXT:F0]      {<diagnostic      description>; <Explanations>}

…

[CONTEXT:F9]      {<diagnostic      description>; <Explanations>}
```

We used concise, list-style generation (beam size = 2, length penalty = 2) with a maximum length sufficient to enumerate all supported codes. Post-processing trims the whitespace and validates the outputs against F0..F9.

For example, given a clinical note describing a patient's experience of auditory hallucinations and social withdrawal, a standard model might struggle to differentiate between schizophrenia and severe depression with psychotic features. With diagnostic descriptions and symptom explanations integrated into the input, the model can anchor its predictions to structured clinical knowledge, thereby improving both precision and robustness.

This enrichment process is especially valuable in the context of auditing psychiatric diagnoses, where the goal is not only to replicate existing clinician labels but also to identify potential misdiagnoses or overlooked conditions. By infusing domain knowledge into the model's input, context enrichment helps mitigate the risks associated with noisy diagnostic labels in the training data and improves the model's capacity to support quality assurance in psychiatric care.

3.4. Model development and tuning

To develop a robust psychiatric diagnosis auditing system, we implemented and evaluated a range of models, spanning traditional ML classifiers, transformer-based multi-label classification models, and a novel QA-based diagnostic prediction approach. The goal was to compare the effectiveness of these different modeling strategies and assess the benefits of task transformation and context enrichment.

Traditional models include logistic regression (Cheng and Hüllermeier, 2009), Naïve Bayes (Zhang et al., 2009), support vector machines (SVM) (de Carvalho and Freitas, 2009), and a neural network classifier (Nam et al., 2014). These models treat psychiatric diagnosis auditing as a multi-label classification task, where each diagnostic label is treated as an independent binary outcome. These baseline models were trained using clinical notes as input and the corresponding psychiatric diagnostic codes as outputs.

Transformer-based models, including bidirectional encoder representations from transformers (BERT) (Devlin et al., 2019), bidirectional and auto-regressive transformers (BART) (Lewis et al., 2019), Llama 3.3 (Grattafiori et al., 2024), BioMistral 7B (Labrak et al., 2024), and Meditron-7B (Chen et al., 2023), were introduced to leverage pretrained language models. BERT was fine-tuned as a multi-label classifier, taking clinical notes as input and simultaneously predicting multiple diagnostic codes. The BART was also fine-tuned in a similar multi-label setup because of its capacity to handle sequence generation tasks, with the goal of comparing its generative capabilities to the conventional classification approach. In addition, Llama 3.3, a state-of-the-art LLM, was evaluated in a zero-shot setting to assess its ability to infer psychiatric diagnoses without task-specific fine-tuning, serving as a benchmark for the effectiveness of domain-adapted transformer models. Biomedical-specific LLMs were also incorporated to assess whether domain-pretraining alone could enhance psychiatric diagnostic auditing. We evaluated BioMistral 7B and Meditron-7B, two biomedical LLMs designed for understanding clinical and biomedical texts. Both models were tested in a zero-shot setting with light prompting to provide a comparison with general-purpose LLMs and the proposed task-specific QA-based models.

QA-based models represent the core methodological innovations of this study. Inspired by recent developments in task reformulation, we reframed the multi-label classification task as a QA problem. Instead of predicting a set of labels independently, the model was prompted with a clinical note along with a structured diagnostic query, such as "what is the psychiatric diagnostic code?" The output was a sequence of diagnostic codes with corresponding descriptions, allowing the model to approach the task as a sequence generation problem. This approach is implemented using a BART-based sequence-to-sequence model.

To enhance the ability of the model to disambiguate complex psychiatric conditions, context enrichment was incorporated into the input. Context enrichment was achieved by appending descriptions of the diagnostic codes (e.g., "F2: schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders") or more detailed diagnostic explanations that linked clinical symptoms to diagnostic categories. This enrichment process aimed to improve the contextual understanding of the model and reduce errors arising from the overlapping nature of psychiatric disorders.

Each model variant was trained using a dataset comprising 48,492 psychiatric patient records. Clinical notes served as the primary input, with the corresponding diagnostic codes as the ground truth output. The models were trained using the hugging face transformers library and the PyTorch framework on graphics processing unit (GPU)-enabled hardware. Hyperparameter tuning was conducted for each model type, including adjustments to batch size, learning rate, and sequence length. The QA models were trained with token-level loss functions, whereas the

traditional and transformer-based classifiers were trained with multi-label cross-entropy loss.

The final set of evaluated models included:

- Traditional models: logistic regression, Naïve Bayes, SVM, and neural network.
- Transformer-based models: BERT, BERT with descriptions, BERT with explanations, and BART with descriptions.
- General-purpose LLM baseline: Llama 3.3 with descriptions, was evaluated in a zero-shot setting to gauge the performance of a general-purpose LLM.
- Biomedical LLM baselines: BioMistral 7B and Meditron-7B, evaluated in a zero-shot setting.
- QA-based models: psychiatric QA model with descriptions, and psychiatric QA model with explanations.

These models were trained and evaluated using multi-label metrics to assess their ability to accurately predict multiple diagnostic codes from clinical notes. The experimental results allow for direct comparisons across approaches, shedding light on the impact of task transformation and context enrichment on diagnostic auditing performance.

3.5. System integration and implementation

The system for psychiatric diagnostic auditing was built upon transformer-based language models, designed to handle the complexity and nuances of unstructured psychiatric clinical notes. A hybrid modeling approach was adopted, combining multi-label classification and QA-based sequence generation techniques to predict diagnostic codes from textual clinical data. The system was developed to process clinical notes, extract relevant diagnostic information, and produce outputs aligned with standardized psychiatric diagnostic terminology.

Central to the system was the BART model, which was selected for its sequence-to-sequence architecture and robust performance in generative tasks such as QA, summarization, and text comprehension. The BART's pretraining process, involving text corruption and reconstruction, was well-suited to the demands of processing unstructured medical texts. Additionally, UnifiedQA, proposed by Khashabi et al. (2020), was incorporated because of its proficiency in QA tasks, having been fine-tuned on a diverse set of QA datasets, equipping it to handle a broad range of diagnostic inquiries. In this study, UnifiedQA was subjected to an additional fine-tuning stage using psychiatric-specific QA pairs, to enhance its capacity to interpret the nuances of mental health diagnoses.

The system was engineered to support both standard multi-label classification tasks and a reformulated QA-based approach. In the QA setting, the system receives a clinical note alongside a structured query, such as, "what is the psychiatric diagnostic code?" The model generates a sequence of diagnostic codes in the response. Context enrichment was integrated into this workflow by appending descriptions of psychiatric codes or explanations linking diagnostic criteria to symptoms, thus providing a model with additional domain knowledge during inference.

The experimental phase involved training and evaluation of a dataset comprising 48,492 clinical records from 7,000 patients. Following data preprocessing, the dataset was split into training (80%), validation (10%), and test (10%) sets, to ensure a robust and balanced evaluation framework. Each clinical note was tokenized and truncated to a maximum input length of 1,024 tokens to align with BART's processing capacity, while the output sequences were constrained to 128 tokens to accommodate multiple diagnostic labels. Beam search decoding with a beam width of 2 and a length penalty of 2 was applied during generation to optimize the diagnostic output quality. All models were trained for up to 16 epochs using the hugging face transformers library and PyTorch with GPU acceleration.

This system integration and implementation process yielded a flexible and context-aware diagnostic auditing tool capable of evaluating psychiatric clinical notes, identifying potential misdiagnoses, and supporting clinicians and auditors in improving the diagnostic accuracy and healthcare quality.

3.6. Evaluation design

Evaluating the performance of models in multi-label psychiatric diagnostic tasks presents unique challenges owing to the complex nature of psychiatric conditions and the potential for overlapping or co-occurring diagnoses. In this study, a comprehensive evaluation framework was adopted to assess the effectiveness of both traditional multi-label classifiers and the proposed QA-based diagnostic auditing models. Given the importance of both precision and recall in diagnostic auditing, multiple evaluation metrics were employed to capture the different aspects of model performance.

First, we introduced a multi-label recall-oriented understudy for gisting evaluation (ROUGE) metric to better align the evaluation with the requirements of the multi-label QA setting. Inspired by the ROUGE score commonly used in text summarization, as proposed by Lin (2004), this custom metric was designed to assess the accuracy and completeness of the generated diagnostic labels of the model. Unlike traditional QA metrics, which often focus on exact matches or independent selections, the multi-label ROUGE metric accounts for the interrelated nature of psychiatric diagnoses, where diagnostic categories frequently overlap. This metric was operationalized using sum-precision, sum-recall, and sum-F1-scores, adapted to evaluate the alignment between the predicted sequence of diagnostic labels and the ground truth diagnoses. Specifically, sum-precision measures the proportion of correctly predicted labels among all predicted labels, sum-recall quantifies the proportion of correctly predicted labels out of all true labels, and Sum-F1 represents their harmonic mean, balancing precision and recall. A worked example demonstrating the calculation of sum-precision, sum-recall, and sum-F1 is provided in Appendix A.

To ensure a robust assessment of diagnostic label prediction across different models and task formulations, additional standard multi-label classification metrics were employed. Micro-precision, micro-recall, and micro-F1-scores treat all predictions equally, emphasizing performance across individual samples, whereas macro-precision, macro-recall, and macro-F1-scores evaluate performance uniformly across all diagnostic labels, as discussed by Santos et al. (2011), highlighting the model's capacity to predict both common and rare conditions. Hamming loss was also incorporated to penalize incorrect predictions, representing the fraction of misclassified labels across all samples (Gao and Zhou, 2011).

To further assess the practical effectiveness of our model, we implemented an automated risk categorization framework and systematically compared the diagnostic conclusions reached by the ML model with those made by treating psychiatrists. This comparison evaluates the level of concordance between the two, identifying the areas of alignment and discrepancies. Based on the degree of agreement observed, each patient's diagnosis was assigned a risk level of high, medium, or low. Importantly, these risk levels are designed to support audit triage rather than to serve as performance metrics or proxies for diagnostic confidence. A high-risk label indicates a significant disparity between the ML model's prediction and the psychiatrist's diagnosis, suggesting potential oversight or the need for further review. Conversely, a low-risk label signifies strong agreement, implying higher confidence in the diagnosis. The medium-risk category captures cases in which partial overlaps exist, reflecting potential ambiguities that warrant further consideration. The risk labeling criteria are listed in Table 2.

To assess the effectiveness of risk categorization, we analyzed the distribution of risk levels across the dataset. By examining the frequency of high-, medium-, and low-risk cases, we aimed to identify potential biases or systematic errors in the model's predictions. This analysis also illustrates how disagreements concentrate differently across partial overlap and fully disjoint diagnostic sets. This analysis provided insights into how well the model replicates psychiatric decision-making and where further refinement may be necessary.

Table 2
Risk labeling for patient diagnosis.

Risk level	Description	Example
Low risk	Assigned when there is a direct match between the machine's output and the doctor's diagnosis, suggesting a high likelihood that the correct diagnosis has been made.	**Doctor:** F2, F3 **Machine:** F2, F3
Medium risk	Allocated in cases where there is a partial match between the deep learning (ML) model's output and the doctor's diagnosis, indicating some level of uncertainty or potential areas for reevaluation.	**Doctor:** F2, F4 **Machine:** F2, F4, F5 **Or** **Doctor:** F1, F4 **Machine:** F4, F5
High risk	Applied when there is a significant mismatch between the machine output and the doctor's diagnosis, raising concerns about the possibility of a misdiagnosis.	**Doctor:** F1, F2, F3 **Machine:** F4, F5

An important consideration in this evaluation was the potential presence of label noise in the training and test data, given that the ground truth diagnostic codes were derived from clinicians' initial assessments, which are susceptible to human error. Label noise is a common challenge in clinical settings, particularly in psychiatry, where diagnoses are often subjective and subject to change. Context enrichment, incorporating diagnostic descriptions and explanations, was introduced to mitigate the impact of noisy labels by providing models with a richer clinical context. While this study primarily relied on recorded diagnoses as the ground truth, future work could incorporate human-in-the-loop validation or weak supervision techniques to further address this limitation.

This evaluation framework ensured that the performance of all models was assessed rigorously and comprehensively, reflecting both the complexity of psychiatric diagnostics and the need for accurate auditing in clinical practice.

Within the evaluation design, risk categories are used exclusively as an audit triage mechanism rather than as a clinical risk score or diagnostic output. They are not used to make, recommend, or alter treatment decisions and are computed offline to prioritize records for differentiated levels of human review. Although both medium- and high-risk cases are flagged for review, they imply distinct audit actions: interpretive reassessment for medium-risk partial overlaps and escalation for high-risk fully disjoint diagnostic sets. Because tiers are heuristic and derived from code-set concordance rather than clinical adjudication, they may yield false positives or negatives and therefore always require clinician judgment. In future work, we will collaborate with psychiatrists to co-design and calibrate thresholds and overrides, and conduct blinded adjudication, strengthening the clinical validity of the triage scheme.

4. Experimental results

This section presents an empirical evaluation of the proposed QA-based psychiatric audit framework. The analysis included (i) descriptive statistics of diagnostic labels, (ii) comparative model performance across traditional and transformer-based methods, (iii) the impact of task transformation (multi-label → QA-based auditing), (iv) the effectiveness of context enrichment, and (v) subgroup analyses to assess fairness across patient demographics. We also include (vi) a brief case study of a medium-risk partial-overlap scenario, illustrating common boundary confusion and showing how the triage scheme flags review without escalation when psychotic features are not documented.

4.1. Descriptive analysis of diagnostic labels

Evaluating the alignment between model predictions and clinician-assigned psychiatric diagnoses is essential for assessing the effectiveness of a model in psychiatric auditing. This study employs a risk categorization framework to systematically analyze the extent to which the model's outputs correspond to clinical diagnoses. Rather than serving as a performance metric, this framework characterizes the structure and severity of disagreement between machine-generated and clinician-assigned diagnostic labels.

The risk categorization distribution reveals the model's diagnostic agreement across three levels:

- Low-risk cases (36.99%): The model's predictions fully matched the psychiatrist's diagnosis, demonstrating strong alignment with clinical assessments.
- Medium-risk cases (45.9%): The model's predictions partially aligned with clinician-assigned diagnoses, indicating some level of diagnostic uncertainty or overlap in symptom interpretation.
- High-risk cases (17.11%): The model's predictions significantly diverge from psychiatric diagnoses, suggesting either misclassification or the identification of alternative conditions not originally recorded by the clinician.

The predominance of medium-risk cases indicates that most disagreements arise from boundary ambiguities rather than safety-critical diagnostic failures. In contrast, high-risk cases, while less frequent, represent qualitatively different disagreements that warrant escalation due to the absence of shared diagnostic grounding.

These results emphasize the importance of context enrichment strategies in auditing psychiatric diagnoses. By integrating diagnostic descriptions and explanations into model inputs, the system can improve the differentiation between similar conditions and enhance interpretability for clinicians. The observed concentration of errors in the medium-risk category further highlights the value of contextual reasoning in resolving clinically adjacent diagnostic confusion. Additionally, analyzing high-risk cases provides valuable insights into systematic model limitations, guiding targeted refinements in training strategies to improve diagnostic reliability.

By employing a structured AI-assisted auditing approach, this study demonstrated the potential of ML models to enhance diagnostic consistency, reduce uncertainty, and provide more transparent, clinically useful psychiatric assessments. Importantly, stratified risk distribution supports the use of differentiated audit actions, enabling scalable triage rather than binary review decisions.

4.2. Overall model performance

The performance of the psychiatric diagnostic models was evaluated using multiple multi-label classification metrics to assess their ability to accurately predict multiple diagnoses for each patient. The results highlight the impact of task transformation (multi-label classification → QA reformulation) and context enrichment (diagnostic descriptions and explanations) on improving the predictive accuracy and reliability.

The baseline models, including the logistic regression, Naïve Bayes, SVM, and neural network classifiers, exhibited moderate performance, with sum-F1-scores ranging from 0.5092 (Naïve Bayes) to 0.7153 (neural network). Among these, SVM performed better than traditional statistical models, achieving a sum-F1-score of 0.6267, but was outperformed by neural and transformer-based models. The neural network classifier (sum-F1: 0.7153) showed the highest performance among the traditional approaches, demonstrating the advantage of deep learning in capturing complex patterns within psychiatric clinical notes.

Transformer-based models, particularly BERT and BART, significantly outperform traditional classifiers. BERT achieved a sum-F1-score of 0.7060, while the BERT model with descriptions (sum-F1: 0.7265) and BERT with explanations (sum-F1: 0.7287) demonstrated the value of context enrichment. The inclusion of diagnostic descriptions and

explanations in the input consistently improved the model performance, confirming that providing structured contextual knowledge aids in the accurate identification of psychiatric conditions.

QA-based models, particularly those employing BART for sequence-to-sequence generation, further enhance diagnostic accuracy. The BART model with descriptions (sum-F1: 0.7226) outperformed traditional classification models, while the psychiatric QA model with descriptions (sum-F1: 0.7244) showed competitive performance. The psychiatric QA model with explanations (sum-F1: 0.7417) emerged as the top-performing model, demonstrating that incorporating both task transformation and enriched contextual inputs leads to the most accurate diagnostic predictions.

The impact of context enrichment was evident across all transformer-based models, particularly in reducing the precision fail rate and hamming loss, both of which are critical for mitigating diagnostic errors. The lowest hamming loss (0.0706) was observed in the BERT with descriptions model, while the psychiatric QA model with explanations achieved the lowest precision failure rate (0.1567), suggesting improved robustness in handling ambiguous cases.

Notably, the LlamA 3.3 model performed significantly worse than most transformer-based baselines, with a sum-F1-score of only 0.3785 and the highest precision failure rate (0.4408). Interestingly, biomedical-pretrained models such as BioMistral 7B (sum-F1: 0.3007) and Meditron-7B (sum-F1: 0.2955) also underperformed, yielding lower accuracy than some traditional classifiers. A key reason for this low performance is the text length limitation of these models, which prevents them from fully leveraging the rich contextual information contained in psychiatric clinical notes. These results suggest that both general-purpose and domain-specific biomedical LLMs struggle with psychiatric auditing tasks without targeted adaptation. By contrast, the proposed psychiatric QA models with contextual enrichment substantially outperformed these baselines, reinforcing the importance of task-specific reformulation and enriched inputs for reliable diagnostic auditing.

The model performance comparisons summarized in Table 3 demonstrate that the QA-based approach, coupled with context enrichment, yields the best results for psychiatric diagnostic auditing. The significant improvements in sum-F1-, micro-F1-, and macro-F1-scores indicate that the proposed methodology enhances the model's ability to disambiguate psychiatric conditions and improve diagnostic reliability.

These results confirm that QA-based psychiatric diagnostic modeling, particularly when augmented with contextual knowledge, outperforms the traditional classification approaches in terms of accuracy, robustness, and interpretability. By reframing multi-label classification as a QA task and integrating diagnostic explanations, the proposed model achieved improved performance compared with BART with descriptions, while reducing errors in ambiguous cases.

4.3. Impact of task transformation (multi-label → QA task)

The transformation of psychiatric diagnosis prediction from a traditional multi-label classification task to a QA-based sequence generation approach resulted in comparable model performance, while providing an alternative formulation for handling complex and overlapping psychiatric conditions. Unlike standard classification models, which independently predict multiple diagnostic labels, the QA-based approach enables the model to generate diagnostic codes in response to structured queries, allowing for greater flexibility and contextual understanding in diagnosis auditing.

The evaluation results highlight the advantages of this transformation. Traditional classification models, such as SVM (sum-F1: 0.6267) and neural network (sum-F1: 0.7153), performed adequately but struggled with diagnostic label ambiguity, often failing to distinguish between highly related conditions, such as Mood [affective] disorders (F3) and intellectual disabilities (F7). In contrast, the QA-based models, particularly psychiatric QA with explanations (sum-F1: 0.7417), consistently outperformed their classification-based counterparts, demonstrating improved recall and more precise differentiation of psychiatric conditions.

One key advantage of QA transformation is its ability to dynamically generate diagnostic sequences rather than treating label prediction as a fixed classification problem. This approach aligns more closely with clinical reasoning, as psychiatric diagnoses are often derived through a structured decision-making process rather than an independent selection of labels. The QA model generates diagnostic outputs in a contextualized manner, allowing it to consider symptom interdependencies and diagnostic criteria before assigning the labels.

Additionally, the precision failure rate was notably lower in the QA models, with psychiatric QA with explanations achieving 0.1567, compared with 0.2845 in SVM and 0.1946 in neural networks. This indicates that the QA approach reduces over-prediction errors, which is particularly important in psychiatric auditing, where unnecessary or incorrect diagnoses can lead to inappropriate treatment plans and increased healthcare costs. Moreover, the hamming loss, which measures incorrect label assignments, was consistently lower in the QA models than in the traditional classifiers, reinforcing the effectiveness of task transformation in mitigating misclassifications.

The improved micro-F1 and macro-F1 scores of the QA-based approach further demonstrated its effectiveness in handling both frequent and less common psychiatric conditions. While traditional classifiers tend to perform well on high-frequency labels but struggle with rare conditions, QA reformulation enables the model to leverage contextual information, resulting in more balanced predictions across diagnostic categories.

Overall, the transition from a multi-label classification task to a QA-based approach enhances the model's ability to capture nuanced

Table 3
Evaluation score of the psychiatric question-anserwing (QA) model.

Model	Sum-precision	Sum-recall	Sum-F1	Precision fail rate	Micro-F1	Macro-F1	Hamming loss
Logistic regression	0.5807	0.4883	0.5156	0.3856	0.62	0.40	0.1044
Naïve Bayes	0.5064	0.5734	0.5092	0.2909	0.54	0.35	0.1686
SVM	0.6588	0.6243	0.6267	0.2845	0.72	0.65	0.0865
Neural network	0.7309	0.7278	0.7153	0.1946	0.77	0.72	0.0782
BERT	0.7317	0.7017	0.7060	0.2237	0.77	0.76	0.0731
BERT with Desc	0.7501	0.7772	0.7265	0.2022	0.72	0.71	0.0706
BERT with Explain	0.7634	0.7221	0.7287	0.1887	0.73	0.72	0.0776
BART with Desc	0.7985	0.6886	0.7226	0.1755	0.73	0.70	0.0839
LLaMA 3.3	0.4459	0.3664	0.3785	0.4408	0.45	0.24	0.1823
BioMistral 7B	0.2083	0.6410	0.3007	0.2249	0.31	0.21	0.4919
Meditron-7B	0.2123	0.5804	0.2955	0.2427	0.30	0.19	0.4248
Psych QA with Desc	0.8029	0.6895	0.7244	0.1711	0.73	0.69	0.0843
Psych QA with Explain	0.8164	0.7087	0.7417	0.1567	0.74	0.70	0.0796

Note: SVM: support vector machine; BERT: bidirectional encoder representations from transformers; BART: bidirectional and auto-regressive transformers; LLaMA: large language model meta AI; Desc: description; Psych: psychiatric.

diagnostic relationships, mitigate label noise, and improve precision in psychiatric diagnosis auditing. These results suggest that reframing diagnostic prediction as a structured QA problem provides a more interpretable and clinically aligned approach, making it a valuable tool for supporting psychiatric quality assurance processes.

4.4. Effectiveness of context enrichment (descriptions and explanations)

The integration of context enrichment, through the addition of diagnostic descriptions and explanations, significantly improved the performance of psychiatric diagnostic models. Context enrichment was designed to address the inherent ambiguity and symptom overlap in psychiatric conditions where similar clinical presentations can lead to misclassification. By supplementing clinical notes with structured diagnostic knowledge, the models were better equipped to differentiate between closely related conditions and improve the accuracy of their predictions.

The evaluation results clearly demonstrate the impact of context enrichment. Across all transformer-based models, those augmented with descriptions and explanations consistently outperform their baseline counterparts. For example, BERT without context enrichment achieved a sum-F1-score of 0.7060, whereas BERT with descriptions improved to 0.7265, and BERT with explanations reached 0.7287. Similarly, in the QA-based models, psychiatric QA with descriptions achieved a sum-F1 of 0.7244, whereas psychiatric QA with explanations improved further to 0.7417, indicating that adding richer diagnostic context enables the model to generate more precise outputs.

The effectiveness of context enrichment was further evident in the precision failure rate and hamming loss. Models with enriched inputs had lower precision fail rates, with psychiatric QA with explanations achieving the lowest rate of 0.1567, compared with 0.2845 for SVM and 0.1946 for standard neural networks. Similarly, the hamming loss was reduced across all enriched models, reinforcing the benefit of providing structured contextual knowledge. These improvements suggest that additional information helps the model reduce incorrect label assignments, particularly in cases where multiple diagnostic categories share overlapping symptoms.

One key advantage of diagnostic descriptions is that they act as anchor points, helping the model to better align its predictions with clinically defined diagnostic categories. For example, without descriptions, a model might struggle to distinguish between F2 (schizophrenia, schizotypal, delusional, and other non-mood psychotic disorders) and F3 (Mood [affective] disorders) due to overlapping symptoms such as delusions or mood disturbances. However, with diagnostic descriptions providing clear symptomatology references, the model can provide more precise classifications.

Explanations further enhance interpretability by linking the symptoms to specific diagnoses. Unlike descriptions that provide static information about diagnostic categories, explanations dynamically connect clinical observations to corresponding psychiatric conditions, reinforcing model decision-making. This added level of contextual reasoning is particularly beneficial in complex cases in which multiple co-occurring conditions exist. For instance, a model without explanations might misclassify a case of anxiety, dissociative, stress-related, somatoform and other nonpsychotic mental disorders (F4) as disorders of adult personality and behavior (F6) due to overlapping anxious symptomatology. However, an explanation-enhanced model, recognizing that OCD is characterized by intrusive thoughts and compulsions, is more likely to accurately distinguish between the two conditions.

The findings suggest that context enrichment, particularly with explanations, improves model generalization and reduces the likelihood of overfitting high-frequency diagnostic patterns. Furthermore, the enhanced interpretability of QA models with explanations makes them more suitable for clinical auditing and decision support because they offer not only a diagnosis but also a rationale for the predicted labels. Overall, the results confirmed that providing structured diagnostic descriptions and symptom-based explanations significantly enhances the accuracy, robustness, and interpretability of psychiatric diagnostic models. This context-aware approach aligns better with clinical reasoning processes, making it a valuable addition to the AI-assisted psychiatric auditing and diagnosis validation systems.

4.5. Subgroup analysis

To evaluate potential systematic biases, subgroup analyses were performed across gender and age categories. Building on the risk-distribution figures (Figs. 2 and 3), we now report the quantitative multi-label performance and misclassification rates by subgroup and provide a short ICD-aware error analysis. Overall, the model maintained broadly consistent performance across genders, while showing more pronounced variation across age groups. This finding suggests general robustness with sensitivity to demographic differences, particularly in certain middle-aged cohorts.

4.5.1. Gender differences

The risk categorization distribution by gender (Fig. 2) revealed distinct patterns in diagnostic concordance. Among male patients, the model assigned a high-risk categorization in 16.76% of cases, medium-risk in 46.02%, and low-risk in 37.22%. For female patients, the high-risk rate was slightly higher (17.38%), medium-risk was the most common (47.23%), and low-risk was lower (35.38%), indicating marginally increased diagnostic ambiguity relative to male patients. Patients without reported gender showed high-risk rate similar to that of males (16.89%), a lower medium-risk rate (40.53%), and the highest low-risk rate (42.58%), suggesting more conservative alignment when gender is unspecified.

These distributional patterns are consistent with the multi-label metrics and error rates listed in Table 4. Across genders, macro-F1 ranged from 0.6598 (female) to 0.7053 (male), and micro-F1 was tightly clustered (0.7390–0.7511), with comparable hamming loss (0.0787–0.017). The misclassification rates were 47.1% (female), 45.1% (male), and 42.4% (not provided), indicating small absolute differences. Taken together, gender effects appear minor and likely reflect modest datasets or practice differences rather than systematic model bias.

4.5.2. Age differences

Age-related patterns in risk categorization (Fig. 3) and multi-label metrics indicated more pronounced disparities. Macro-F1 varied substantially from 0.4407 (60–69) to 0.7591 (40–49), a range of 0.3184, whereas micro-F1 ranged from 0.6980 (30–39) to 0.8222 (≥70), a difference of 0.124 (Table 4). Error rates were highest in the 60–69 group (misclassification rate, 52.0%) and lowest in the ≥70 group (29.6%). The 40–49 group showed the strongest balance (macro-F1 0.7591; micro-F1 0.7674), with a relatively low hamming loss (0.075). These patterns are consistent with (i) greater diagnostic complexity and comorbidity in certain middle-aged cohorts, (ii) potential data imbalance across age brackets, and (iii) overlapping symptom profiles that are

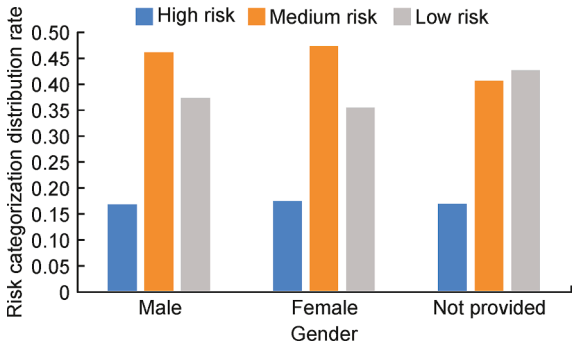


Fig. 2. Subgroup analysis by gender.

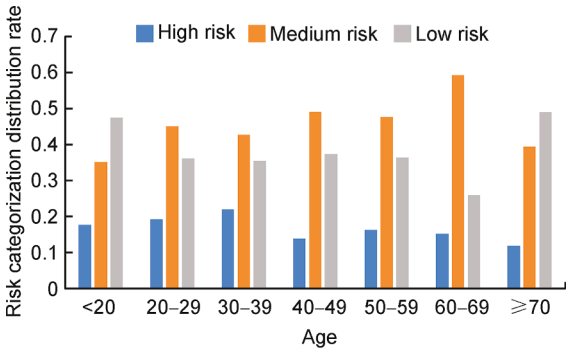


Fig. 3. Subgroup analysis by age.

difficult to separate in multi-label settings. Although expert review is needed to fully validate these interpretations, the observed trends suggest that age-related variation may reflect underlying data and clinical factors, not solely model error.

4.5.3. Short error analysis (ICD-aware)

The most frequent true predicted confusions were F4 → F2 (493 cases), F4 → F3 (334), F4 → F1 (296), F3 → F1 (262), F3 → F2 (206), F1 → F3 (155), F2 → F1 (150), F3 → F4 (133), F9 → F3 (124), and F4 → F0 (117). Consistent with the risk definitions.

- **Medium risk (partial overlap):** Bidirectional confusion among clinically adjacent families (e.g., F3 ↔F4; F1 ↔F3; F2 ↔F3) typically reflects shared symptom clusters and is most often categorized as medium, since at least one correct family is present.
- **High risk (mismatches):** Moves from F4 to more distant families without any overlap with clinician labels (e.g., F4 → F0 when the doctor labeled F1/F3/F4; or multi-family shifts that omit all clinician labels) correspond to the high definition (disjoint sets).

The most frequently missed true labels were F4 (1016 cases), F3 (573), F1 (300), F9 (292), F2 (260), and F0 (152), with smaller counts for F5, F6, and F8. The most common false positives were F1 (407), F3 (269), F2 (170), F4 (129), and F0 (94). These distributions reinforce that most disagreements arise at clinically porous boundaries (anxiety/stress-related, mood, psychotic, substance-related), aligning with medium-risk, partial-overlap behavior; comparatively fewer cases manifest as fully disjoint sets indicative of high risk.

4.5.4. Interpretation and implications

Taken together with the subgroup results, these findings indicate that label-level disagreements are concentrated in clinically adjacent ICD families (F3–F4–F2–F1), in which the risk framework classifies them primarily as medium rather than high. Importantly, the absence of critical FNs demonstrates that safety-relevant recall is maintained even when boundary cases remain a challenge. In future work, targeted augmentation for the most frequent confusion pairs (e.g., contrastive examples for F3 vs. F4 and F2 vs. F3), along with demographic-aware fine-tuning, may reduce medium-risk ambiguity while preserving the safety-critical performance.

4.6. Case study: partial diagnostic overlap—medium risk

4.6.1. Case

Individual office progress note documenting worsening problems with moods. Mental status: well groomed, appropriate attire, average eye contact; affect depressed and anxious; behavior cooperative; suicidal and homicidal ideation denied; speech not recorded; insight and judgment fair. Plan indicates continued services; patient reports numbness on Topamax, prefers to keep dosage stable, sleeping better on medications, and reports current disability until January. In our label framework, the clinician's labels were F2 and F4, while the model predicted F3 and F4. Risk classification by our scheme: medium risk, because both sets include F4 but diverge on the second family (clinician: F2; model: F3).

4.6.2. Analysis

The narrative emphasizes anxiety and depressed affect with no documented psychotic features, which supports the shared F4 label and makes the model's additional F3 (mood) plausible. The clinician's inclusion of F2 (psychotic spectrum) was not substantiated in the note excerpt, suggesting either information not captured in this note or potential label noise. This pattern mirrors the frequent boundary confusion observed in our error analysis for F3–F4–F2. Because there is an overlap on F4 and no safety-critical content was missed, this instance is best interpreted as a medium-risk discrepancy that merits review rather than a high-risk failure.

4.7. Case study: disjoint diagnostic sets—high risk

4.7.1. Case

Individual office progress note documenting ongoing psychosocial stressors and treatment engagement. Mental status: well groomed, appropriate attire, average eye contact; affect appropriate; behavior cooperative; speech normal; insight and judgment good; suicidal and homicidal ideation denied. Client reported refocusing energy on areas of control, including building his resume, resource development, looking for an internship, and speaking with an immigration lawyer at school. Client described a desire to gain social confidence to overcome loneliness and reported rumination when alone interfering with joy. cognitive behavioural therapy was used to explore patterns, interferences, and solutions. Plan indicates continued services; client will continue journaling worries to process solutions and seek increased social integration through activities such as going to the gym and volunteer groups. In our label framework, the clinician's labels were F1 (mental and behavioral disorders due to psychoactive substance use) and F3 (mood [affective] disorders), while the model predicted F2 (schizophrenia, schizotypal,

Table 4
Subgroup performance and error rates in the multi-label setting.

Subgroup		Cases	Hamming loss	Macro-F1	Micro-F1	Misclassification rate (%)
Gender	Female	2,600	0.0817	0.6598	0.7390	47.1
	Male	1,569	0.0787	0.7053	0.7453	45.1
	Not provided	681	0.0733	0.7021	0.7511	42.4
Age	< 20	414	0.0790	0.5809	0.7212	43.0
	20–29	1,110	0.0839	0.5923	0.7275	48.3
	30–39	842	0.0939	0.6696	0.6980	50.0
	40–49	879	0.0750	0.7591	0.7674	44.0
	50–59	761	0.0739	0.5865	0.7619	44.3
	60–69	506	0.0822	0.4407	0.7432	52.0
	⩾70	338	0.0512	0.5669	0.8222	29.6

delusional, and other non-mood psychotic disorders). Risk classification by our scheme: high risk.

4.7.2. Analysis

This case constitutes a high-risk discrepancy because the clinician-assigned labels (F1 and F3) and the model prediction (F2) belong to non-overlapping diagnostic families with distinct symptom profiles and clinical implications. The narrative centers on loneliness, rumination, reduced social confidence, and goal-directed behavior, which are more consistent with affective or adjustment-related concerns than substance-related pathology or psychotic symptomatology. Notably, the note contains no documentation of intoxication, withdrawal, substance misuse, hallucinations, delusions, disorganized thoughts, or impaired reality testing. Unlike medium-risk cases, where partial overlap reflects boundary ambiguity among clinically adjacent categories, this instance reflects a fundamental divergence in the diagnostic interpretation. Because the disagreement spans affective, substance-related, and psychotic spectra without shared evidentiary grounding in the documented clinical text, the case is appropriately classified as high-risk and would warrant escalation for focused clinical adjudication rather than routine secondary review.

5. Discussion

5.1. Key findings and implications

This study demonstrates the effectiveness of task transformation (multi-label classification → QA-based approach) and context enrichment (detailed descriptions and explanations) in enhancing psychiatric diagnosis auditing. The experimental results confirmed that the proposed psychiatric QA model with explanations achieved the highest sum-F1-score (0.7417) and the lowest precision fail rate (0.1567), significantly outperforming traditional classification models. These findings underscore the advantages of reconfiguring psychiatric diagnosis prediction as a structured QA task, which enables context-aware, flexible diagnostic reasoning and reduces misclassification errors.

One of the most significant improvements observed in this study was the impact of context enrichment on the model performance. Models augmented with diagnostic descriptions and explanations consistently outperformed their baseline counterparts, highlighting the importance of incorporating structured clinical knowledge into input representations. In particular, explanations provided a clear advantage over descriptions alone, as they dynamically linked symptomatology with diagnostic categories, reducing ambiguity in psychiatric condition classification. This finding suggests that contextual reasoning, rather than simple pattern recognition, is critical for accurate AI-assisted diagnostic auditing.

The risk categorization framework provides further insight into model reliability. Approximately 36.99% of the cases were classified as low-risk, indicating strong agreement between the model's predictions and professional psychiatric diagnoses. 45.9% of the cases fell into the medium-risk category, where the model's predictions partially overlapped with clinical assessments, highlighting potential areas for further refinement. These medium-risk cases represent the most frequent and analytically informative audit scenarios because they are surface boundary conditions and documentation gaps that are not captured by aggregate accuracy metrics. The 17.11% of the cases categorized as high-risk revealed instances where model outputs significantly diverged from clinician diagnoses, signaling the need for improvements in handling complex psychiatric conditions with overlapping symptom profiles. Unlike medium-risk cases, high-risk discrepancies reflect fully disjoint diagnostic interpretations and therefore warrant escalation rather than routine secondary review.

Subgroup analysis showed broadly consistent performance across gender groups, while more pronounced variations were observed across age groups. However, the slightly higher high-risk categorization rates among female patients suggest potential areas where model adjustments could improve diagnostic precision across demographic subgroups. Additionally, the 60–69 age group showed the highest misclassification rate, highlighting the need for further model refinement across age-related subgroups.

These findings have significant implications for clinical decision-making support and psychiatric auditing. The QA-based approach offers a more interpretable and context-aware method for psychiatric diagnosis validation, making it a promising tool for quality assurance in mental healthcare and medical insurance verification. Importantly, three-level risk stratification enables differentiated audit actions, supporting a scalable triage rather than a binary review/no-review workflow. By reducing diagnostic inconsistencies and providing structured explanations for predictions, AI-driven auditing systems could help psychiatrists identify potential misdiagnoses, flag ambiguous cases for review, and ensure that insurance claims are based on accurate and well-substantiated diagnoses. Future studies should focus on further refining demographic-sensitive model tuning, expanding context enrichment strategies, and integrating human-in-the-loop feedback mechanisms to enhance interpretability and trustworthiness in psychiatric AI applications.

A fuller discussion of label noise, data heterogeneity, calibration/uncertainty, and deployment considerations with concrete mitigation steps is presented in Section 6.

5.2. Theoretical contributions

This study advances the field of AI-assisted psychiatric diagnostics by introducing a novel task transformation approach, reframing multi-label psychiatric diagnosis prediction as a QA task. Unlike traditional classification models that assign diagnostic labels independently, the proposed QA-based method mimics the structured reasoning process used in clinical decision-making, allowing for more flexible and context-aware predictions. This transformation represents a paradigm shift in how ML models approach psychiatric auditing, offering a framework that aligns better with human diagnostic workflows.

Another key theoretical contribution is the integration of context enrichment into the prediction of psychiatric diagnoses. By incorporating diagnostic descriptions and explanations, this study demonstrated that providing structured clinical knowledge during inference improves diagnostic accuracy and interpretability. While prior research on medical AI has focused on classification-based models, this study highlights the importance of contextual reasoning in psychiatric diagnosis, showing that models perform better when supplemented by domain-specific knowledge. The results indicate that explanations offer a greater performance boost than descriptions alone, reinforcing the idea that AI models benefit not only from additional information but also from structured, explanatory reasoning that links symptoms to diagnostic labels.

This research also contributes to the evaluation of psychiatric diagnostic models by proposing a risk categorization framework that systematically compares ML-generated diagnoses with clinician-assigned labels to identify areas of alignment and discrepancy. Unlike prior works, where disagreement is treated as an undifferentiated error, the proposed low/medium/high stratification operationalizes disagreement into analytically distinct categories with different audit meanings. Rather than serving as a proxy for predictive accuracy, the risk framework functions as an information-systems mechanism for audit triage and governance. This framework has broader implications for AI auditing in medical settings, as it establishes a structured approach for identifying misdiagnoses, flagging cases for further review, and enhancing the DSS.

Additionally, this study extends the use of multi-label evaluation metrics by adapting ROUGE-based summarization techniques to psychiatric diagnosis prediction. Traditional classification metrics often fail to account for semantic relationships and label interdependencies in

patients with psychiatric conditions. The proposed multi-label ROUGE metric provides a more holistic measure of model performance, capturing the structured nature of psychiatric diagnoses and offering an alternative to conventional multi-label evaluation techniques.

Overall, these contributions advance the intersection of AI, clinical psychiatry, and medical auditing, demonstrating that task reconfiguration, context enrichment, and structured evaluation frameworks can enhance the reliability and interpretability of AI-driven psychiatric diagnostic systems. By framing disagreement as a first-class object of analysis rather than a residual error term, this work opens new directions for AI-based auditing research that emphasizes governance, accountability, and human–AI collaboration. These findings open new directions for future research on medical AI, emphasizing the need for models that go beyond pattern recognition to incorporate clinical reasoning, contextual awareness, and robust quality assurance mechanisms.

5.3. Practical implications for healthcare and insurance auditing

The findings of this study have significant implications for the real-world application of AI in psychiatric auditing, particularly for enhancing diagnostic consistency, clinical decision support, and medical insurance verification. Psychiatric diagnoses are inherently complex and subjective, often relying on clinician judgment rather than on definitive biomarkers. This variability creates challenges for quality assurance in mental health care, where misdiagnoses, inconsistencies, and diagnostic ambiguity can lead to inappropriate treatments, increased healthcare costs, and insurance claim disputes. The proposed QA-based AI system with context enrichment addresses these challenges by systematically auditing clinician-assigned diagnoses, identifying discrepancies, and providing structured explanations, thus making it a valuable tool for both clinical and administrative auditing workflows.

The primary application of this framework is medical auditing and insurance claim validation. Psychiatric diagnoses often determine insurance coverage, disability claims, and reimbursements. Errors or inconsistencies in recorded diagnoses can lead to claim denials, unnecessary investigations, or legal disputes, which increase administrative burdens for both providers and insurers. By integrating AI-assisted auditing into review processes, medical auditors can cross-check clinician-assigned diagnoses against AI-evaluated evidence, surfacing potential misclassifications or unsupported claims before they escalate. The risk categorization framework further strengthens this capability by flagging medium- and high-risk cases for targeted human review, ensuring higher reliability and consistency in psychiatric evaluation and claims adjudication.

Beyond insurance auditing, the proposed system has practical value in clinical quality control and second-opinion workflow. Hospitals and psychiatric clinics could adopt this tool to flag cases where diagnostic uncertainty is high, prompting peer review or additional assessments. This application is particularly relevant for training psychiatric residents, where AI-generated explanations and evidence highlights can support learning by revealing sources of inconsistency and reinforcing best-practice diagnostic reasoning.

The framework also has implications in legal and forensic psychiatry, in which psychiatric diagnoses influence the determination of competency, responsibility, or custody. In such settings, AI-assisted auditing can provide structured, evidence-linked evaluations that support fairness, transparency, and reliability in high-stake decision-making.

At a broader level, hospitals, regulators, and insurers could leverage such systems to establish standardized protocols for psychiatric auditing, reduce subjectivity and mitigate bias in mental health assessments. Integrating real-time auditing modules within EHR platforms would allow clinicians to receive immediate feedback on the consistency and evidence-basis of their diagnostic entries, improve reliability and reduce downstream errors.

In summary, by improving diagnostic transparency, standardization, and quality assurance, AI-assisted psychiatric auditing has the potential to reduce healthcare costs, improve patient outcomes, and strengthen trust in psychiatric decision-making support. It offers a scalable pathway to bridge critical gaps in psychiatric care, insurance adjudication, and governance, positioning auditing as a central component of responsible AI adoption in healthcare.

5.4. Ethical considerations

The integration of AI into psychiatric diagnosis auditing raises important ethical considerations, particularly regarding bias, transparency, and human oversight in decision-making. Given that psychiatric assessments are inherently subjective and influenced by cultural, social, and clinical factors, there is a risk that AI models may reinforce the existing biases present in the training data. If certain demographic groups are underrepresented or misclassified in historical diagnostic records, AI-driven auditing tools can perpetuate disparities in mental health assessments. Ensuring fairness and inclusivity in AI models requires continuous monitoring, bias mitigation techniques, and diverse training datasets that reflect the full spectrum of psychiatric conditions across populations.

Transparency is another critical concern because AI-based auditing systems must be interpretable and explainable by clinicians and medical reviewers. Black-box models that produce diagnostic evaluations without clear reasoning can erode trust in AI-assisted auditing tools, making it difficult for healthcare professionals to validate AI-generated recommendations. By incorporating context enrichment strategies, such as diagnostic explanations, this study takes a step toward enhancing model interpretability. However, further efforts are required to ensure that AI-generated insights remain clinically meaningful and understandable to practitioners.

Despite advancements in AI-assisted psychiatric auditing, human oversight remains essential to ensure that AI recommendations are used as a supporting tool rather than a definitive authority. Psychiatric diagnoses often involve nuanced clinical judgments, patient history, and evolving symptom presentations, which an AI model may not fully capture. Therefore, AI-based auditing systems should be designed to augment rather than replace human expertise, allowing psychiatrists to critically evaluate AI-generated outputs and make informed final decisions.

Regulatory guidelines, ethical frameworks, and institutional review mechanisms must be established to responsibly implement AI in psychiatric auditing. These frameworks should outline the best practices for AI deployment in clinical settings, including data privacy protection, continuous performance evaluation, and mechanisms for addressing AI-detected discrepancies in diagnoses. Ultimately, while AI holds great promise for improving diagnostic accuracy, auditing efficiency, and healthcare quality, its ethical deployment requires a careful balancing of technological capabilities with clinical judgment, patient well-being, and fairness in mental healthcare.

6. Limitations and future research

6.1. Limitations of the study

This study had several limitations. The dataset comes from a single psychiatric clinic and combines optical character recognition (OCR)'d and native EHR notes with heterogeneous templates and provider styles. Such variability introduces potential distribution shift and may limit generalizability. We did not conduct a multi-site external validation. Clinician-entered diagnoses can be noisy, temporally unstable, and influenced by billing/documentation constraints. We did not apply explicit noise-modeling techniques, although context enrichment may reduce some ambiguity. Our evaluation emphasized classification-style

metrics and did not assess confidence calibration or implement abstention when uncertain—factors were important for audit triage.

Moreover, while descriptions and explanations improved transparency, we did not test explanation faithfulness or obtain clinician ratings of the explanatory quality. The risk tiers used here are heuristic, tuned on a development split without psychiatrist co-design, and intended strictly for audit triage rather than for clinical risk scores. Fairness analyses were limited to age and gender; we did not examine intersectional subgroups or conduct formal fairness audits. Finally, this was a retrospective study; we did not evaluate shadow deployment, user burden, or downstream effects within live clinical workflows.

6.2. Future research

Future studies will focus on noise-aware learning and label refinement. We plan to incorporate loss correction and early learning regularization, explore co-teaching strategies, and use weak supervision (rules and treatment proxies) together with confident learning to surface likely mislabeled samples. A human-in-the-loop adjudication protocol will target stratified medium/high triage cases to improve ground truth quality and quantify agreement.

To improve the robustness under heterogeneity and distribution shift, we will continue pretraining/fine-tuning on local clinical corpora, use section-aware encoders and layout cues, and evaluate domain-adversarial or group-robust optimization. We will add explicit drift detection with a scheduled recalibration. External validity will be assessed via multi-institution studies and temporal holdout tests, reporting performance, calibration, and error profiles by site and time.

We will strengthen calibration and uncertainty handling by reporting Brier scores and expected calibration errors alongside F1, adding selective prediction (abstaining when uncertain), and exploring conformal prediction to produce diagnosis sets with coverage guarantees that are particularly valuable for audit triage. For interpretability, we will add span-level evidence highlighting, run faithfulness tests, and collect clinician ratings of explanatory usefulness and fidelity. We also plan to systematically evaluate and visualize variations in reasoning depth and evidence presentation, quantifying how different models balance explanation richness with conciseness. Such analyses will help to link explanation patterns to clinicians' trust and adoption.

To ensure clinically grounded risk tiers, we will co-design thresholds, override with psychiatrists, and conduct blinded adjudication with inter-rater agreement targets. In parallel, we will expand fairness analyses beyond age and gender, adopt reweighting or thresholding if disparities are detected, publish model cards and data statements, and maintain audit logs and verification for traceability.

To further enhance generalizability, we will extend the proposed QA-based framework to long context biomedical-pretrained models and optimize the LLMs. This will allow a systematic evaluation of whether domain pretraining or advanced reasoning mechanisms provide additional gains when combined with task reformulation and context enrichment.

Finally, we will pursue a prospective, workflow-centered evaluation through shadow deployment within EHRs, measuring reviewer effort, turnaround time, and escalation rates, and study how audit flags influence documentation and claims resolution. Operational safeguards will include fairness audits with subgroup reporting, reviewer-facing outputs only, abstention with escalation to human experts, comprehensive logging of prompts, and evaluation of stronger privacy controls. We will also develop integration patterns for EHR systems with clear abstinence pathways and periodic governance reviews to reassess bias, drift, and safety.

7. Conclusion

This study demonstrates the effectiveness of AI-assisted psychiatric auditing through task transformation (a multi-label classification → QA-based approach) and context enrichment (diagnostic descriptions and explanations). By systematically comparing ML-generated diagnoses with clinician assessments, the proposed model enhances diagnostic accuracy and identifies cases requiring further scrutiny, reinforcing its value as a decision-support tool in mental healthcare.

The proposed framework demonstrates potential for future integration into clinical auditing workflows and represents a step forward in AI-driven psychiatric diagnostics, with potential benefits including improved diagnostic consistency, reduced clinician workload, and enhanced medical auditing processes. By flagging high-risk cases and providing structured explanations, this system can help psychiatrists identify potential misdiagnoses and refine diagnostic accuracy, ultimately improving patient outcomes.

Beyond its clinical implications, this study highlights important ethical and practical considerations in AI deployment, particularly in terms of bias mitigation, data privacy, and transparency. These findings emphasize the need for continuous monitoring, human oversight, and integration strategies to ensure that AI tools remain reliable, interpretable, and equitable across diverse patient populations.

This study contributes to the growing field of AI in medical diagnostics by offering a novel approach that combines ML efficiency with human expertise. By bridging AI and clinical decision-making, this research paves the way for future advancements in AI-driven auditing, ensuring that technology serves as a complementary tool to enhance, rather than replace, psychiatric professionals in delivering high-quality mental healthcare.

CRediT authorship contribution statement

William (Hongmin) Du: Writing – original draft, Resources, Data curation, Conceptualization. **Guanghua Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis. **Jiahui (Norah) Xu:** Writing – original draft, Data curation. **Miklos A. Vasarhelyi:** Supervision, Resources, Conceptualization.

Ethics statement

This study did not involve any clinical, pharmacological, biological, or medical procedures on human or animal subjects. The research was based exclusively on the analysis of anonymized secondary data and did not include any interaction with human participants. In accordance with institutional guidelines, ethical review and approval were not required for this study. Written informed consent was not required because all data were fully anonymized prior to analysis.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this study, we used ChatGPT (OpenAI) to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed, and took full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.dsm.2026.01.002.

References

Abd-Alrazaq, A., AlSaad, R., Alhuwail, D., et al., 2023. Large language models in medical education: opportunities, challenges, and future directions. JMIR Med. Educ. 9 (Jun.), e48291.

Abdel-Karim, B., Pfeuffer, N., Carl, K.V., et al., 2023. How AI-based systems can induce reflections: the case of AI-augmented diagnostic work. MIS Q. 47 (4), 1395–1424.

Adam, G., Rampášek, L., Safikhani, Z., et al., 2020. Machine learning approaches to drug response prediction: challenges and recent progress. npj Precis. Oncol. 4 (Jun.), 19.

Aftab, A., Banicki, K., Ruffalo, M.L., et al., 2024. Psychiatric Diagnosis: A Clinical Guide to Navigating Diagnostic Pluralism. J. Nerv. Ment. Dis. 212 (8), 445–454.

Aggarwal, A., Tam, C.C., Wu, D., et al., 2023. Artificial intelligence-based chatbots for promoting health behavioral changes: systematic review. J. Med. Internet Res. 25 (Feb.), e40789.

Allsopp, K., Read, J., Corcoran, R., et al., 2019. Heterogeneity in psychiatric diagnostic classification. Psychiatry Res. 279 (Sep.), 15–22.

Aminabee, S., 2024. The future of healthcare and patient-centric care: digital innovations, trends, and predictions. In: Emerging Technologies for Health Literacy and Medical Practice. IGI Global Scientific Publishing, pp. 240–262.

Amirahmadi, A., Ohlsson, M., Etminani, K., 2023. Deep learning prediction models based on EHR trajectories: a systematic review. J. Biomed. Inform. 144 (Aug.), 104430.

Badawy, M., Ramadan, N., Hefny, H.A., 2023. Healthcare predictive analytics using machine learning and deep learning techniques: a survey. J. Electr. Syst. Inf. Technol. 10 (1), 40.

Bagheri, M., Bagheritabar, M., Alizadeh, S., et al., 2025. Machine-learning-powered information systems: a systematic literature review for developing multi-objective healthcare management. Appl. Sci. 15 (1), 296.

Berner, E.S., Graber, M.L., 2008. Overconfidence as a cause of diagnostic error in medicine. Am. J. Med. 121 (5 Suppl), S2–S23.

Bordini, B.J., Stephany, A., Kliegman, R., 2017. Overcoming diagnostic errors in medical practice. J. Pediatr. 185 (Jun.), 19–25.e1.

Bosomworth, N.J., 2011. Practical use of the Framingham risk score in primary prevention: Canadian perspective. Can. Fam. Physician 57 (4), 417–423.

Bradford, A., Meyer, A.N.D., Khan, S., et al., 2024. Diagnostic error in mental health: a review. BMJ Qual. Saf. 33 (10), 663–672.

Burton-Jones, A., Volkoff, O., 2017. How can we develop contextualized theories of effective use? A demonstration in the context of community-care electronic health records. Inf. Syst. Res. 28 (3), 468–489.

Bzdok, D., Meyer-Lindenberg, A., 2018. Machine learning for precision psychiatry: opportunities and challenges. Biol. Psychiatry Cogn. Neurosci. Neuroimaging 3 (3), 223–230.

Chaturvedi, S., 2025. Impact of diagnostic uncertainty in modern psychiatry. Physician 9 (3), 1.

Chekroud, A.M., Bondar, J., Delgadillo, J., et al., 2021. The promise of machine learning in predicting treatment outcomes in psychiatry. World Psychiatry 20 (2), 154–170.

Chen, Z., Hernández-Cano, A., Romanou, A., et al., 2023. MEDITRON-70B: Scaling Medical Pretraining for Large Language Models arXiv:2311.16079v1.

Cheng, W., Hüllermeier, E., 2009. Combining instance-based learning and logistic regression for multilabel classification. Mach. Learn. 76 (2), 211–225.

Clark, L.A., Cuthbert, B., Lewis-Fernández, R., et al., 2017. Three approaches to understanding and classifying mental disorder: ICD-11, DSM-5, and the national institute of mental health's research domain criteria (RDoC). Psychol. Sci. Publ. Interest 18 (2), 72–145.

Dai, H.J., Su, C.H., Lee, Y.Q., et al., 2021. Deep learning-based natural language processing for screening psychiatric patients. Front. Psychiatr. 11 (Jan.), 533949.

Dakanalis, A., Wiederhold, B.K., Riva, G., 2024. Artificial intelligence: a game-changer for mental health care. Cyberpsychol. Behav. Soc. Netw. 27 (2), 100–104.

de Carvalho, A.C.P.L.F., Freitas, A.A., 2009. A tutorial on multi-label classification techniques. In: Foundations of Computational Intelligence Volume 5. Springer, Berlin, Heidelberg, pp. 177–195.

Devlin, J., Chang, M.W., Lee, K., et al., 2019. BERT: pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). ACL, pp. 4171–4186.

Dwyer, D.B., Falkai, P., Koutsouleris, N., 2018. Machine learning approaches for clinical psychology and psychiatry. Annu. Rev. Clin. Psychol. 14 (May), 91–118.

Efremov, A., 2025. Psychiatry in the context of changing cultural norms: mental disorders among migrants and refugees. J. Behav. Cogn. Ther. 35 (4), 100544.

Evans, S.R., Pennello, G., Pantoja-Galicia, N., et al., 2016. Benefit-risk evaluation for diagnostics: a framework (BED-FRAME). Clin. Infect. Dis. 63 (6), 812–817.

Gao, W., Zhou, Z.H., 2011. On the consistency of multi-label learning. In: Proceedings of the 24th Annual Conference on Learning Theory, pp. 341–358.

García-Gutiérrez, M.S., Navarrete, F., Sala, F., et al., 2020. Biomarkers in psychiatry: concept, definition, types and relevance to the clinical reality. Front. Psychiatr. 11 (May), 432.

Gibson, J., Atkins, D.C., Creed, T.A., et al., 2022. Multi-label multi-task deep learning for behavioral coding. IEEE Trans. Affect. Comput. 13 (1), 508–518.

Grattafiori, A., Dubey, A., Jauhri, A., et al., 2024. The Llama 3 Herd of Models arXiv: 2407.21783.

Guo, T., Bardhan, I.R., Ding, Y., et al., 2024. An explainable artificial intelligence approach using graph learning to predict intensive care unit length of stay. Inf. Syst. Res. 36 (3), 1478–1501.

Jackson, R.G., Patel, R., Jayatilleke, N., et al., 2017. Natural language processing to extract symptoms of severe mental illness from clinical text: the Clinical Record Interactive Search Comprehensive Data Extraction (CRIS-CODE) project. BMJ Open 7 (1), e012012.

Javaid, M., Haleem, A., Pratap Singh, R., et al., 2022. Significance of machine learning in healthcare: features, pillars and applications. Int. J. Intell. Netw. 3 (Jan.), 58–73.

Jayamani, J., Janardan, C.C., Appan, S.V., et al., 2022. A practical tool for risk management in clinical laboratories. Cureus 14 (12), e32774.

Jussupow, E., Spohrer, K., Heinzl, A., et al., 2021. Augmenting medical diagnosis decisions? An investigation into physicians' decision-making process with artificial intelligence. Inf. Syst. Res. 32 (3), 713–735.

Keramatian, K., Levit, A., 2023. Caring for youth with co-occurring substance use and severe psychiatric disorders: diagnostic challenges and clinical implications. J. Can. Acad. Child Adolesc. Psychiatry 32 (3), 202.

Khashabi, D., Min, S., Khot, T., et al., 2020. UnifiedQA: Crossing Format Boundaries with a Single QA System arXiv: 2005.00700.

Konjevod, M., Sáiz, J., Bordoy, L., et al., 2025. Validated metabolomic biomarkers in psychiatric disorders: a narrative review. Mol. Med. 31 (1), 254.

Kourounis, G., Elmahmudi, A.A., Thomson, B., et al., 2023. Computer image analysis with artificial intelligence: a practical introduction to convolutional neural networks for medical professionals. Postgrad. Med. J. 99 (1178), 1287–1294.

Labrak, Y., Bazoge, A., Morin, E., et al., 2024. BioMistral: a collection of opensource pretrained large language models for medical domains. In: Findings of the Association for Computational Linguistics: ACL 2024. ACL, pp. 5848–5864.

Lee, C.T., Palacios, J., Richards, D., et al., 2023. The precision in psychiatry (PIP) study: testing an Internet-based methodology for accelerating research in treatment prediction and personalisation. BMC Psychiatry 23 (1), 25.

Lewis, M., Liu, Y., Goyal, N., et al., 2019. BART: Denoising Sequence-To-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 7871–7880.

Li, Z., Liu, F., Yang, W., et al., 2022. A survey of convolutional neural networks: analysis, applications, and prospects. IEEE Trans. Neural Netw. Learn. Syst. 33 (12), 6999–7019.

Lin, C.Y., 2004. ROUGE: a package for automatic evaluation of summaries. In: Text Summarization Branches Out. ACL, pp. 74–81.

Malik, H., Farooq, M.S., Khelifi, A., et al., 2020. A comparison of transfer learning performance versus health experts in disease diagnosis from medical imaging. IEEE Access 8 (Jun.), 139367–139386.

Marey, A., Arjmand, P., Alerab, A.D.S., et al., 2024. Explainability, transparency and black box challenges of AI in radiology: impact on patient care in cardiovascular radiology. Egypt. J. Radiol. Nucl. Med. 55 (1), 183.

McGorry, P., Keshavan, M., Goldstone, S., et al., 2014. Biomarkers and clinical staging in psychiatry. World Psychiatry 13 (3), 211–223.

McGowan, J., Wojahn, A., Nicolini, J.R., 2023. Risk Management Event Evaluation and Responsibilities. StatPearls Publishing, Treasure Island.

Meyer, A.N.D., Giardina, T.D., Khawaja, L., et al., 2021. Patient and clinician experiences of uncertainty in the diagnostic process: current understanding and future directions. Patient Educ. Couns. 104 (11), 2606–2615.

Nam, J., Kim, J., Loza Mencía, E., et al., 2014. Large-scale multi-label text classification: revisiting neural networks. In: Machine Learning and Knowledge Discovery in Databases. Springer, Berlin, Heidelberg, pp. 437–452.

Njoroge, S.W., Nichols, J.H., 2014. Risk management in the clinical laboratory. Ann. Lab. Med. 34 (4), 274–278.

Omar, M., Soffer, S., Charney, A.W., et al., 2024. Applications of large language models in psychiatry: a systematic review. Front. Psychiatr. 15 (Jun.), 1422807.

Park, D.J., Park, M.W., Lee, H., et al., 2021. Development of machine learning model for diagnostic disease prediction based on laboratory tests. Sci. Rep. 11 (Apr.), 7567.

Ren, J., Liu, N., Wu, X., 2020. Clinical questionnaire filling based on question answering framework. Int. J. Med. Inform. 141 (Sep.), 104225.

Rix, K., 2026. DSM and ICD classifications in medico-legal reporting: misperceptions, misunderstandings and misuse. BJPsych Adv. 32 (1), 44–56.

Rosen, R.D., Sapra, A., 2022. TNM classification. In: StatPearls [Internet]. StatPearls Publishing. Treasure Island.

Rutter, M., 2011. Research review: child psychiatric diagnosis and classification: concepts, findings, challenges and potential. J. Child Psychol. Psychiatry 52 (6), 647–660.

Santos, A.M., Canuto, A.M., Neto, A.F., 2011. A comparative analysis of classification methods to multi-label tasks in different application domains. Int. J. Comput. Inf. Syst. Ind. Manag. Appl. 3 (Nov.), 10.

Saoud, R., Abou Heidar, N., Cimadamore, A., et al., 2020. Incorporating prognostic biomarkers into risk assessment models and TNM staging for prostate cancer. Cells 9 (9), 2116.

Sbodio, M.L., López, V., Hoang, T.L., et al., 2024. Collaborative artificial intelligence system for investigation of healthcare claims compliance. Sci. Rep. 14 (1), 11884.

Sharma, N., Kaushik, P., 2025. Integration of AI in healthcare systems—a discussion of the challenges and opportunities of integrating AI in healthcare systems for disease detection and diagnosis. In: AI in Disease Detection: Advancements and Applications, pp. 239–263.

Stein, D.J., Shoptaw, S.J., Vigo, D.V., et al., 2022. Psychiatric diagnosis and treatment in the 21st century: paradigm shifts versus incremental integration. World Psychiatry 21 (3), 393–414.

Sun, J., Dong, Q.X., Wang, S.W., et al., 2023. Artificial intelligence in psychiatry research, diagnosis, and therapy. Asian J. Psychiatry 87 (Sep.), 103705.

Ten Cate, O., Regehr, G., 2019. The power of subjectivity in the assessment of medical trainees. Acad. Med. 94 (3), 333–337.

Thakkar, A., Gupta, A., De Sousa, A., 2024. Artificial intelligence in positive mental health: a narrative review. Front. Digit. Health 6 (Mar.), 1280235.

Vasconcelos-Moreno, M.P., Pinto, J.V., Hasse-Sousa, M., et al., 2025. Diagnosis, clinical features, differential diagnosis, and psychiatric comorbidities. In: Bipolar Disorder: an Evidence-Based Clinical Guide. Springer, pp. 105–146.
Wang, Y., Hu, B., Zhao, Y., et al., 2021. Applications of system dynamics models in chronic disease prevention: a systematic review. Prev. Chronic Dis. 18 (Dec.), 210175.
Wang, J., Rao, C., Goh, M., et al., 2023. Risk assessment of coronary heart disease based on cloud-random forest. Artif. Intell. Rev. 56 (1), 203–232.
Wang, Z., Li, H., Huang, D., et al., 2025. HealthQ: unveiling questioning capabilities of LLM chains in healthcare conversations. Smart Health 36 (Jun.), 100570.
Wu, H., Deng, Z., Evans, R., 2022. Building patients' trust in psychologists in online mental health communities. Data Sci. Manag. 5 (Mar.), 21–27.
Xing, Y., 2025. Regulatory Alignment: Does Medical Necessity Standardization Affect Healthcare Provision? Available at SSRN 5351261.
Zhang, M.L., Peña, J.M., Robles, V., 2009. Feature selection for multi-label naive Bayes classification. Inf. Sci. 179 (19), 3218–3229.
Zhang, A., Xing, L., Zou, J., et al., 2022. Shifting machine learning for healthcare from development to deployment and from models to data. Nat. Biomed. Eng. 6 (12), 1330–1345.